

A maintenance optimization model for mission-oriented systems based on Wiener degradation

Chiming Guo^a, Wenbin Wang^{b,*}, Bo Guo^a, Xiaosheng Si^c

^a College of Information Systems and Management, National University of Defense Technology, Changsha, China

^b Dongling School of Economics and Management, University of Science and Technology, Beijing, China

^c Department of Automation, Xi'an Institute of High-Tech, Xi'an, China

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ABSTRACT

Over the past few decades, condition-based maintenance (CBM) has attracted many researchers because of its effectiveness and practical significance. This paper deals with mission-oriented systems subject to gradual degradation modeled by a Wiener stochastic process within the context of CBM. For a mission-oriented system, the mission usually has constraints on availability/reliability, the opportunity for maintenance actions, and the monitoring type (continuous or discrete). Furthermore, in practice, a mission-oriented system may undertake some preventive maintenance (PM) and after such PM, the system may return to an intermediate state between an as-good-as new state and an as-bad-as old state, i.e., the PM is not perfect and only partially restores the system. However, very few CBM models integrated these mission constraints together with an imperfect nature of the PM into the course of optimizing the PM policy. This paper develops a model to optimize the PM policy in terms of the maintenance related cost jointly considering the mission constraints and the imperfect PM nature. A numerical example is presented to demonstrate the proposed model. The comparison with the simulated results and the sensitivity analysis show the usefulness of the optimization model for mission-oriented system maintenance presented in this paper.

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1. Introduction

A mission-oriented system is a system that is required to perform certain missions with fixed durations at various times in its lifetime [1]. Examples of such systems can be found in the ground systems of the aerospace launch center such as the fuel feed system; and in the military systems such as the avionics portion of an airborne weapon system. If the system fails during the mission phase, the mission must be aborted. A mission-oriented system is usually monitored by condition monitoring after each mission, and if the system is found to be in an unacceptable state, some PM will be carried out, which may return the system to an intermediate state between an as-good-as new state and an as-bad-as old state, i.e., the PM is not perfect and only partially restores the system. In addition, a mission can have constraints on availability/reliability, the opportunity for PM actions, and the monitoring type (continuous or discrete). The system will eventually be replaced by a new one when the following events occur: the system fails; the system availability/reliability does not satisfy the mission requirement or the system service time reaches a pre-specified number of years.

This pre-specified number of years of service is usually determined by the design of the system or other factors such as safety, obsolescence etc.

Like other systems, many mission-oriented systems suffer deterioration in operation. When the deterioration of the system reaches a failure threshold, the system should be repaired or replaced. If the degradation of a mission-oriented system can be measured and modeled appropriately, we can take maintenance actions on the basis of the observed degradation data before the system has a failure. This is what the condition-based maintenance (CBM) aimed for, which is to achieve a cost-effective maintenance based on the system condition information. Over the past few decades, CBM has attracted many researchers' interest because of its effectiveness and practical significance. For a comprehensive literature review in CBM, see Scarf [2], Wang [3], Jardine et al. [4], van Noortwijk [5]. The decisions made from CBM depend on the system condition data obtained by monitoring techniques among others. The framework of CBM mainly includes three parts, i.e., condition data acquisition, reliability and/or lifetime estimation from condition monitoring data, and CBM decision-making [4]. Degradation modeling is a way to estimate the remaining life time based on monitored data and it is commonly used when degradation measure is available.

Degradation modeling provides an important means for maintenance decision-making [6]. Many types of degradation models

* Corresponding author. Tel.: +86 10 62332744.

E-mail address: wangwb@ustb.edu.cn (W. Wang).

Nomenclature

Acronyms

CBM	condition-based maintenance
PM	preventive maintenance
CM	corrective maintenance
pdf	probability density function
cdf	cumulative distribution function
RUL	remaining useful life

Notations

$D(t)$	degradation state at time t with realization d_t
l_a	preventive threshold corresponding to perform a PM
l_f	failure threshold corresponding to perform a CM
T_f or T_a	first-passage time to threshold l_f or l_a
C_i	cost per inspection
C_P	cost per unit time for PM
C_R	cost per replacement

C_C	cost per CM
τ	fixed mission time
U	maximum mission number before a renewal
W_j	period of the j th renewal cycle
$T_i^{(j)}$	inter-maintenance time between the $(i-1)$ th PM and the i th PM in the j th renewal cycle
M_i	duration for the i th imperfect PM
ξ	duration for the replacement due to a CM action
W_j	mission availability threshold
$A1[i]$	mission availability for the i th maintenance cycle
$A2$	achieved long run availability for the system
S_i	residual damage after the i th PM
$E[S_i]$	expectation of S_i
$T_{op(1)}, T_{op(2)}, T_{op(3)}$	expected operating time for three different types of renewal cycles
$T_{nop(1)}, T_{nop(2)}, T_{nop(3)}$	expected non-operating time for three different types of renewal cycles
$Cost_{(1)}, Cost_{(2)}, Cost_{(3)}$	expected cost for three different types of renewal cycles.

have been developed in the literature. General path models and stochastic process models are two kinds of frequently used models to describe the degradation processes [7]. A general path model focuses on the inter-item variability and thus can be used to estimate the lifetime distribution for the population, see Lu and Meeker [8], Park and Bae [9]. However, the degradation uncertainty for an individual is not taken into account in this kind of models. A stochastic process model focuses on the individual item behavior and can remedy the shortage of the general path model [10].

Stochastic-process-based models such as Markov chains [11], Gamma processes [12,13] and Wiener processes [14,15] have been widely used to model degradation processes, and have shown advantages over the general path model in CBM modeling. For a comprehensive review, see Si et al. [6]. Among these stochastic models, a Wiener process model can describe a non-monotonic degradation process with Gaussian noise, and provide a good description for some system behavior [16]. Crowder and Lawless [17] incorporated a random effect into the Wiener process to specify the random variation among the individuals and proved that CBM can offer much cost saving over a failure replacement policy. Whitmore and Schenkelberg [18] used a Wiener process with a time scale transformation to model the performance degradation of the self-regulating heating cable. The degradation data collected from sensors usually do not show a monotonic trend due to the noise influence. By regarding every degradation increment as an additive superposition of a large number of small effects, the degradation process can be assumed to follow a normal distribution. Therefore, in this paper, a Wiener process is chosen to describe mission-oriented system degradation.

Since the deterioration for a mission-oriented systems is described by a degradation model such as a Wiener process, the degradation threshold to initiate the PM (called the PM threshold) is the decision variable constituting the PM policy. In order to optimize the PM policy, some criteria, such as cost, availability and risk, are usually chosen as the optimization objectives in practice. Marseguerra et al. [19] used Genetic Algorithm and Monte Carlo simulation to optimize CBM with two objectives including profit and availability. Grall et al. [20] assumed a replacement policy with a multi-level control-limit rule to optimize maintenance policy based on a cost criterion. Further, Castanier et al. [21] assumed a repair/replacement policy based on a multi-level control limit rule and obtained the optimal

thresholds based on both the cost and availability criteria. Amari and McLaughlin [22] utilized a Markov chain to describe a deterioration system subject to periodic inspection for the CBM model. The optimal inspection frequency and the maintenance threshold were found to maximize the system availability. Liao et al. [23] investigated a realistic maintenance policy which achieved the maximum availability level for the continuously degrading system. Li et al. [24] developed a maintenance model with reliability constraints for the mission-oriented avionic system based on the lifetime distribution. However, very few CBM models integrated the mission constraints together with an imperfect nature of the PM into the course of optimizing the PM policy with an objective function of the related maintenance cost.

The main objective of this paper is to develop a model to optimize the PM policy for a mission-oriented system based on a Wiener degradation process so that the mission requirements can be achieved with a minimum cost. Compared with the similar studies by Liao et al. [23] and Li et al. [24], the proposed model in this paper has the following innovations: (a) considering the mission constraints to PM occasions; (b) involving the imperfect PM effects and (c) taking the system availability requirement imposed by the mission into account.

Specifically, we consider a system with a mission availability constraint subject to periodic PM that is imperfect. For modeling the PM process, we use a geometric process for the residual damage and an exponential function for the mean PM duration, respectively. The residual damage is the left-over damage uncovered by the imperfect PM. Through classifying the system lifecycle, we present a probabilistic model for the PM process. An enumeration algorithm is used to find the optimal PM threshold. By analyzing the results, it is found that ignoring the mission constraints can lead to unsatisfactory results for a mission-oriented system.

The remaining parts are organized as follows. In Section 2, the system degradation is described by a Wiener stochastic process model, and the remaining useful life (RUL) distribution is obtained. Section 3 is devoted to describe the PM policy of the mission-oriented system. In Section 4, the influences of imperfect PM are modeled. In Section 5, the system renewal process is classified and a probabilistic analysis is presented in order to establish the optimization model. Section 6 formulates the optimization model and provides the optimization procedure. Section 7 provides a numerical example to illustrate the proposed

model and validates usefulness of the proposed model. Conclusions are drawn in Section 8.

2. Degradation model based on a Wiener process

2.1. Wiener stochastic process

Let the system degradation at time t be represented by $D(t)$ with realization d_t . The degradation process $\{D(t): t \in \mathbf{R}_+\}$ is assumed to follow a Wiener process and is expressed by

$$D(t) = \mu t + \sigma W(t), \quad \mu \geq 0 \quad (1)$$

where $W(t)$ is a standard Brownian motion, μ is the slope of the linear drift and $\sigma > 0$ is the diffusion coefficient. The Wiener process is not monotonously increasing but the mean degradation is linearly increasing in t , i.e., $E(D(t)|\mu) = \mu t$.

$D(t)$ has the following properties [25]:

- $D(0) = 0$ almost surely.
- For any time sequence $\{t_q\}_{1 \leq q \leq n, q \in \mathbf{Z}^+}$ with $0 < t_1 < t_2 < \dots < t_n$, the random increments $D(t_1)$, $D(t_2 - t_1)$, $\dots$, $D(t_n - t_{n-1})$, where $D(t_{j+1} - t_j) = D(t_{j+1}) - D(t_j)$, are independent, and any $D(t) - D(r) (t > r > 0)$ follows $N(0, \sigma^2 | t - r|)$.
- The paths of $D(t)$ are continuous with probability one.

2.2. Remaining useful life distribution

The probability distribution of RUL is very important for CBM since the RUL estimation can be used to schedule the PM action [26]. One popular definition of the RUL using Wiener process modeling is the first passage time which is the time from now to the point that the degradation first reaches a critical failure threshold l_f . Such first-passage time T_f can be defined as $T_f = \inf\{t | D(t) \geq l_f, t \geq 0\}$ at time zero. T_f has an inverse Gaussian distribution with probability density function (pdf) [18,27]

$$f_{T_f}(t) = \frac{l_f}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(l_f - \mu t)^2}{2\sigma^2 t}\right) \quad (2)$$

The corresponding cumulative distribution function (cdf) is

$$F_{T_f}(t) = P(T_f < t) = \Phi\left(\frac{-l_f + \mu t}{\sigma\sqrt{t}}\right) + \exp\left(\frac{2\mu l_f}{\sigma^2}\right) \Phi\left(\frac{-l_f - \mu t}{\sigma\sqrt{t}}\right) \quad (3)$$

Based on the failure threshold, the RUL T_ω at state $D(\omega) = d_\omega < l_f$ can be established as

$$T_\omega = \inf\{v \geq 0 | D(v + \omega) \geq l_f\} \text{ if } D(\omega) < l_f; \text{ otherwise } T_\omega = 0.$$

According to the independent increments property, and the homogeneous Markov property, we have

$$\begin{aligned} T_\omega &= \inf\{v \geq 0 | D(v + \omega) - D(\omega) \geq l_f - D(\omega)\} \\ &= \inf\{v \geq 0 | D(v) \geq l_f - D(\omega)\}. \end{aligned} \quad (4)$$

Similar to the first-passage time distribution from time zero, the remaining lifetime T_ω knowing $D(\omega)$ also follows an inverse Gaussian distribution. The conditional pdf of T_ω knowing that $D(\omega) = d_\omega$ can be obtained by replacing l_f with $l_f - d_\omega$ in (2) as

$$f_{T_\omega}(t | d_\omega) = \frac{l_f - d_\omega}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(l_f - d_\omega - \mu t)^2}{2\sigma^2 t}\right) \quad (5)$$

And the conditional cdf of the remaining life time knowing that $D(\omega) = d_\omega$ is

$$F_{T_\omega}(t | d_\omega) = \Phi\left(\frac{\mu t - (l_f - d_\omega)}{\sigma\sqrt{t}}\right) + \exp\left(\frac{2\mu(l_f - d_\omega)}{\sigma^2}\right) \Phi\left(\frac{-(l_f - d_\omega) - \mu t}{\sigma\sqrt{t}}\right) \quad (6)$$

3. PM policy for a mission oriented system

A mission-oriented system usually works intermittently with required reliability/availability constraints by the mission, such as military systems, aerospace shuttles. Many researchers assumed that the PM action can be carried out immediately if the system degradation state crosses an alarm threshold. However, for most mission-oriented systems, PM may not be able to be carried out during the mission. Furthermore the PM is usually imperfect, i.e., the PM action does not bring the system to the as-good-as-new state. In order to develop a realistic PM model for the mission-oriented system, we propose the following modeling assumptions.

3.1. Assumptions

- Each mission duration is a constant denoted by τ . The system is monitored perfectly and periodically at the end of the mission and a fixed cost C_l is incurred. The time needed for inspection is negligible since it is usually small compared with the mission time.
- The PM is imperfect and does not restore the system to the as-good-as new condition.
- If the system state satisfies $l_a \leq D(t) < l_f$ at the end of a mission, a PM is carried out immediately before the next mission and the cost per unit time for the PM is C_p . The PM threshold l_a is a decision variable, $l_a > 0$.
- The expectation of the duration for each imperfect PM action shows an increasing trend with the severity of deterioration increasing.
- If $D(t) \geq l_f$ occurs during a mission, the mission fails. A corrective maintenance (CM) is carried out immediately to recover the system to be as good as new and a fixed cost C_c is incurred. This is called a system renewal by CM.
- If $D(t) < l_a$ occurs at the end of a mission, the system is left unchanged.
- When the system operating time reaches $U\tau$, a preventive replacement is carried out to restore the system to an as-good-as new condition and a cost C_R is incurred. We also use $U \in \mathbf{Z}^+$ which is the maximum number of missions that any system can carry out.
- Before system failure or $U\tau$, the system can also be replaced when the system availability constraint is violated. This will incur the same C_R . Both non-CM replacements are called renewals by preventive replacement, though we will classify them further later.
- All renewals consume an equal amount of time.
- The cost satisfies $C_l < C_p < C_R < C_c$. This is because CM will include the penalty cost due to mission failure which could be very high and certainly higher than the PM costs. The preventive replacement cost at the end of the life of the system will cost more than the PM cost during the life of the system because of Assumption 4. Inspection cost is the smallest among them as expected since it only involves some checking for degradation without any repair or maintenance action.

3.2. PM process description

Fig. 1 illustrates the system state evolution with PMs and renewals. In Fig. 1 $\{W_j\}_{j \in \mathbf{N}}$ is a renewal sequence, i.e., W_j denotes the j th renewal cycle length; $T_i^{(j)}$ denotes the i th inter-PM time in the j th renewal cycle; $M_i^{(j)}$ denotes the duration occupied by the i th PM in the j th renewal cycle. From Fig. 1, we can see that $T_2^{(1)}$ is the second operating period after $R_1^{(1)}$. A renewal duration is

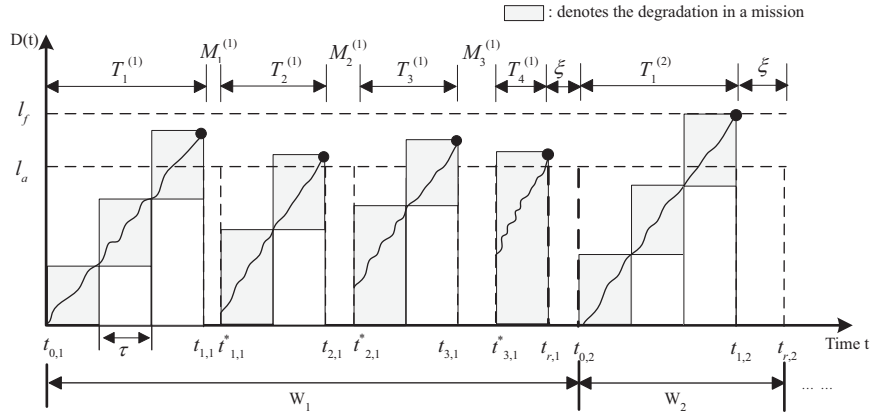


Fig. 1. The system PM processes and renewal cycles.

denoted by ζ . $t_{i,j}$ and $t_{i,j}^*$ denote the initial time epoch and the end time epoch for the i th PM action in the j th renewal cycle, respectively. $t_{0,j}$ denotes the initial time epoch for the first operating period in the j th renewal cycle. $t_{r,j}$ denotes the initial time epoch for the j th renewal action.

In practice, because of the imperfect PM and system ageing, the degradation will start from some value non-zero after each PM and stochastically increasing with the sequence of PMs in each renewal cycle. This value is called the residual damage and will shorten the inter-PM interval and prolong the duration required for the next PM [28]. As shown in Fig. 1, the inter-PM intervals between the imperfect PM actions show a decreasing trend, i.e., $E[T_1^{(1)}] > E[T_2^{(1)}] > E[T_3^{(1)}] > E[T_4^{(1)}]$ and the lengths of PM durations show an increasing trend with the sequence of PMs, i.e., $E[M_1^{(1)}] < E[M_2^{(1)}] < E[M_3^{(1)}]$.

Two kinds of availability are defined referring to the work of Liao et al. [23], denoted by $A1[i]$ and $A2$, respectively. $A1[i]$ is the average mission availability for the i th PM cycle, where the i th PM cycle is composed of the i th inter-PM interval and the $(i-1)$ th PM duration. In each renewal cycle, the inter-PM interval and the PM duration will be re-indexed as $T[i]$ and $M[i-1]$ to denote the i th inter-PM interval and the $(i-1)$ th PM duration for the i th PM cycle, respectively. The mission availability for the i th PM cycle is

$$A1[i] = \frac{E[T[i]]}{E[T[i]] + E[M[i-1]]} \quad (7)$$

For the mission-oriented system, if the mission availability for a PM cycle within the fixed lifetime $U\tau$, is reduced to an unacceptable level ζ , a replacement will be carried out to restore the system to a new one.

$A2$ is the achieved long run availability in a renewal cycle, and the downtime includes the time of PM, CM and replacement. The average long run availability for a renewal cycle is

$$A2 = \frac{\text{Totaluptime/Cycle}}{\text{Totaluptime/cycle} + \text{Total downtime/cycle}} \quad (8)$$

4. The imperfect PM model and its influences

In order to optimize the availability, it is necessary to investigate the influences of the imperfect PM. According to the description of the imperfect PM characteristics in Section 3, the residual damage model and the required PM duration model are given as follows.

4.1. The residual damage model

In general, the mean of the residual damage after each PM will exhibit an increasing trend with the sequence of PMs, i.e., the ability for the PM action to improve the system health state is weakening.

The residual damage after each imperfect PM in the j th renewal cycle, $D(t_{i,j}^*)$, falls randomly in the interval $[0, l_a]$. With Y_i defined as $Y_i = Y(t_{i,j}^*) = l_a - D(t_{i,j}^*)$ and for simplicity it is assumed that $\{Y_i, i = 1, 2, \dots\}$ follows an exponential-based stochastic decreasing geometric process with $Y_{i+1} \leq_{st} Y_i$. The cdf and pdf of Y_i are

$$G_i(y) = \begin{cases} c_i (1 - \exp(-a^{i-1} \frac{\varepsilon}{l_a} y)) & y \in [0, l_a] \\ 1 & y \in (l_a, +\infty) \\ 0 & y \in (-\infty, 0), \end{cases} \quad (9)$$

$$g_i(y) = \begin{cases} c_i a^{i-1} \frac{\varepsilon}{l_a} \exp(-a^{i-1} \frac{\varepsilon}{l_a} y) & y \in [0, l_a] \\ 0 & \text{others,} \end{cases} \quad (10)$$

where $a > 1$ is the ratio of the geometric process, $\varepsilon > 0$ is a factor to describe the scale parameter of the exponential distribution and $c_i = 1/(1 - \exp(-\varepsilon a^{i-1}))$, $i = 1, 2, \dots, n$.

For the sake of simplicity, S_i is used to denote $D(t_{i,j}^*)$. By variable replacement, the cdf and pdf of residual damage S_i are

$$H_i(s) = \begin{cases} c_i \exp(-\varepsilon a^{i-1}) (\exp(a^{i-1} \frac{\varepsilon}{l_a} s) - 1) & s \in [0, l_a] \\ 0 & s \in (-\infty, 0) \\ 1 & s \in (l_a, +\infty), \end{cases} \quad (11)$$

$$h_i(s) = \begin{cases} c_i a^{i-1} \frac{\varepsilon}{l_a} \exp(-a^{i-1} \frac{\varepsilon}{l_a} (l_a - s)) & s \in [0, l_a] \\ 0 & \text{others,} \end{cases} \quad (12)$$

where s is the realization of S_i and $s \in [0, l_a]$.

The expectation of S_i is

$$E[S_i] = \int_{-\infty}^{+\infty} sh_i(s)ds = \int_0^{l_a} sh_i(s)ds = c_i l_a \left[1 - \frac{1}{\varepsilon a^{i-1}} (1 - \exp(-\varepsilon a^{i-1})) \right] \quad (13)$$

with $S_0 = 0$.

It can be proved that $E[S_i]$ is an increasing function of the number of imperfect PMs, as shown in Appendix A based on Eqs. (11)–(13). Of course other distributions such as Weibull can also be used for Eqs. (9) and (10), but we cannot prove analytically that $E[S_i]$ increases with the number of PMs. So to keep the consistence of the paper, we limit our attention to the exponential case in this paper. The PM cycle number cannot be infinite due to

either a failure or the operating time reaching $U\tau$ or the availability constraint because of the accumulated residual damages, and therefore the number of missions is always limited in a renewal cycle.

4.2. The PM duration model

In reality, the durations required to perform each PM action in a renewal cycle usually show an increasing trend, since a longer time is needed to maintain the severely degraded system than a slightly degraded one. For a general situation, the PM durations can be assumed to be independent but not identically distributed random variables, but in this paper we only use the mean of such duration for simplicity. With more mathematical manipulation, we can use a distribution rather than the mean, but it is not our main focus here.

Let M_i denote the duration required for the i th PM action, and for any $i \neq j$, M_i is independent of M_j . In model building, the expectation of M_i is desired to present an increasing trend with the PM threshold l_a and the PM sequence number. Similar to the choice in [23], we set

$$E[M_i] = i\gamma l_a \exp(i\lambda l_a) \quad (14)$$

where $\gamma > 0$, $\lambda > 0$, $i = 1, 2, \dots$ and $M_0 = 0$. Assume the parameters, γ and λ , to be independent of l_a , and can be estimated by the PM duration data with statistical methods.

4.3. The influences of the imperfect PM

The characteristics of an imperfect PM, i.e., the stochastic increasing residual damage and the increasing trend of the PM duration, will affect the average mission availability and the reliability for each inter-PM interval in a renewal cycle. It can be proved that the average mission availability for a PM cycle $[t_{ij}, t_{i+1,j}]$, will decrease with the PM sequence number.

Lemma 1. For the mission-oriented system assumed in this paper, the average mission availability in a PM cycle, $A1(i)$, has a decreasing trend with the imperfect PM sequence number.

Proof. According to the average mission availability definition,

$$\begin{aligned} A1(i) &= \frac{E[T_i]}{E[T_i] + E[M_{i-1}]} = \frac{E[E[T_i|S_{i-1}]]}{E[E[T_i|S_{i-1}]] + E[M_{i-1}]} \\ &= \frac{1}{1 + E[M_{i-1}]/E[E[T_i|S_{i-1}]]} \end{aligned} \quad (15)$$

where $E[T_i|S_{i-1}]$ is the expected inter-PM interval conditional on S_{i-1} . Denote T_a as the first-passage time to l_a , $T_a = \inf\{t|D(t) \geq l_a, t \geq 0\}$. According to (5) and (6), the pdf and cdf of T_a conditional on the residual damage S_{i-1} are

$$f_{T_a}(t|S_{i-1}) = \frac{l_a - S_{i-1}}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(l_a - S_{i-1} - \mu t)^2}{2\sigma^2 t}\right) \quad (16)$$

$$\begin{aligned} F_{T_a}(t|S_{i-1}) &= P(T_a < t|S_{i-1}) = \Phi\left(\frac{-(l_a - S_{i-1}) + \mu t}{\sigma\sqrt{t}}\right) \\ &\quad + \exp\left(\frac{2\mu(l_a - S_{i-1})}{\sigma^2}\right) \Phi\left(\frac{-(l_a - S_{i-1}) - \mu t}{\sigma\sqrt{t}}\right) \end{aligned} \quad (17)$$

So the conditional expectation of T_i is

$$\begin{aligned} E[T_i|S_{i-1}] &= \int_0^{+\infty} t f_{T_a}(t|S_{i-1}) dt = \int_0^{+\infty} t \frac{l_a - S_{i-1}}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(l_a - S_{i-1} - \mu t)^2}{2\sigma^2 t}\right) \\ &\quad dt = \frac{l_a - S_{i-1}}{\mu} \end{aligned} \quad (18)$$

Furthermore, we have

$$E[E[T_i|S_{i-1}]] = \int_0^{l_a} \frac{l_a - s}{\mu} h_{i-1}(s) ds = \frac{l_a - E[S_{i-1}]}{\mu} \quad (19)$$

According to (13), $E[S_{i-1}]$ is positively correlated with the imperfect PM sequence number, so $E[E[T_i|S_{i-1}]]$ is a decreasing function with the PM sequence number. It means that the expectation of T_i is decreasing with the PM sequence number i .

Furthermore, $E[M_i]$ is an increasing function of the imperfect PM sequence number based on (14). Obviously, $E[M_i]/E[E[T_i|S_{i-1}]]$ is an increasing function with the PM sequence number. Therefore, $A1(i)$ shows a decreasing trend as the imperfect PM sequence number increases, i.e., $A1(0) > A1(1) > A1(2) > \dots > A1(N-1)$. $\square$

Based on the properties of the Wiener degradation process, we can also find that the reliability in an inter-PM interval decreases with the increased residual damage.

Lemma 2. For a Wiener-degradation system with a linear drift, the reliability over a fixed time period decreases with the residual damage.

Proof. Assuming the initial degradation, i.e., the residual damage is s , the reliability over the fixed time period ϑ can be defined as $R(\vartheta|s, l_f) = P(T_f > \vartheta|s, s < l_f)$ (20)

According to (6), we can obtain

$$R(\vartheta|s, l_f) = 1 - \Phi\left(\frac{\mu\vartheta - (l_f - s)}{\sigma\sqrt{\vartheta}}\right) - \exp\left(\frac{2\mu(l_f - s)}{\sigma^2}\right) \Phi\left(\frac{-(l_f - s) - \mu\vartheta}{\sigma\sqrt{\vartheta}}\right) \quad (21)$$

From (21), it can be seen that the reliability over a fixed time period θ with the residual damage s and failure threshold l_f is equal to the reliability over a fixed time θ with the initial degradation $D_0 = 0$ and failure threshold $l_f - s$.

Defining two residual damage states $s1, s2$ and $0 < s1 < s2 < l_f$, equivalently, $R(\vartheta|s1, l_f)$ and $R(\vartheta|s2, l_f)$ can be transformed to $R(\theta|L1)$ and $R(\theta|L2)$ without residual damage, where $L1 = l_f - s1$ and $L2 = l_f - s2$, $L1 > L2 > 0$.

Denoting $A(\vartheta) = \max_{0 \leq v \leq \vartheta} \{D(v)\}$, we define two sets $Q1$ and $Q2$, $Q1 = \{\text{All type 1 Events} | \text{type 1} : A(\vartheta) < L1\}$, (22)

$$Q2 = \{\text{All type 2 Events} | \text{type 2} : A(\vartheta) < L2\} \quad (23)$$

$Q1$ is the event set for all the possible events that satisfy $A(\vartheta) < L1$ and $Q2$ is the set for all the possible events that satisfy $A(\vartheta) < L2$. Then we have

$$R(\vartheta|L1) = P(A(\vartheta) < L1) = \sum_{E \in Q1} P(E) \quad (24)$$

$$R(\vartheta|L2) = P(A(\vartheta) < L2) = \sum_{E \in Q2} P(E) \quad (25)$$

where $P(E)$ is the probability for the occurrence of event E .

As $Q2 \subset Q1$, it can be concluded that $R(\vartheta|L1) > R(\vartheta|L2)$.

Thus it is proved that the reliability over a fixed time period decreases with the residual damage. $\square$

Lemma 1. shows that the average mission availability in a PM cycle decreases with the number of the PM sequence. **Lemma 2** proves that the failure opportunity increases with the increased residual damage.

In order to keep the system operating normally under the mission requirement, PM can be used to improve the system availability. However, the number of PM cycles could not be infinite. From **Lemma 1**, it can be seen that the imperfect PM will lead the system availability to an unacceptable level eventually.

In Section 6, an optimization model will be established to explore the optimal PM threshold under the assumption of a limited PM cycle number before a renewal.

5. Maintained system evolution

To seek the optimal PM policy needs to model the system evolution process. Renewal process techniques provide efficient tools for such tasks. In order to make the model tractable, the expectation of the residual damage is adopted to study the maintained system evolution, and it provides a meaningful way to study the stochastic characteristics for a mission-oriented system.

5.1. Probability for different maintenance types

The evolution of $D(t)_{t \geq 0}$ is a renewal process with the regeneration times either by a CM or a preventive renewal. Within the i th PM cycle, there exist three possible scenarios for the κ th mission.

- 1) If the system state crosses l_f within the κ th mission, a CM renewal will be carried out. The expected probability for such an event with respect to S_{i-1} is

$$\begin{aligned} E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) + S_{i-1} \geq l_f)] \\ = \int_0^{l_a} P(D((\kappa-1)\tau) + s < l_a \cap D(\kappa\tau) + s \geq l_f) h_{i-1}(s) ds \\ = \int_0^{l_a} \left((1 - F_{T_a}((\kappa-1)\tau | s)) \int_0^{l_a-s} F_{l_f-x}(\tau; x | s) f_{\phi}(x; (\kappa-1)\tau) dx \right) h_{i-1}(s) ds, \end{aligned} \quad (26)$$

where

$$\begin{aligned} F_{T_a}(t | s) = P(T_a < t | s) \\ = \Phi\left(\frac{-(l_a-s) + \mu t}{\sigma\sqrt{t}}\right) + \exp\left(\frac{2\mu(l_a-s)}{\sigma^2}\right) \Phi\left(\frac{-(l_a-s) - \mu t}{\sigma\sqrt{t}}\right), \end{aligned} \quad (27)$$

$$\begin{aligned} F_{l_f-x}(\theta; x | s) = P(T_{l_f-x} < \theta | s) = \Phi\left(\frac{-(l_f-x-s) + \mu\theta}{\sigma\sqrt{\theta}}\right) \\ + \exp\left(\frac{2\mu(l_f-x-s)}{\sigma^2}\right) \Phi\left(\frac{-(l_f-x-s) - \mu\theta}{\sigma\sqrt{\theta}}\right), \end{aligned} \quad (28)$$

$$f_{\phi}(x; t) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{(x-\mu t)^2}{2\sigma^2 t}\right) \quad (29)$$

(29) is the pdf of the system state $D(t)=x$ at time t [29]. The detailed computation can be found in Appendix B-(1).

- 2) If the system state satisfies $D((\kappa-1)\tau) < l_a$ and $l_a \leq D(\kappa\tau) < l_f$, a PM will be carried out after the κ th mission, the expected probability for such an event with respect to S_{i-1} is

$$\begin{aligned} E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap l_a \leq D(\kappa\tau) + S_{i-1} < l_f)] \\ = \int_0^{l_a} P(D((\kappa-1)\tau) + s < l_a \cap l_a \leq D(\kappa\tau) + s < l_f) h_{i-1}(s) ds \\ = \int_0^{l_a} ((1 - F_{T_a}((\kappa-1)\tau | s)) \int_0^{l_a-s} (F_{l_a-x}(\tau; x | s) - F_{l_f-x}(\tau; x | s)) \\ f_{\phi}(x; (\kappa-1)\tau) dx) h_{i-1}(s) ds, \end{aligned} \quad (30)$$

where

$$\begin{aligned} F_{l_a-x}(\theta; x | s) = P(T_{l_a-x} < \theta | s) = \Phi\left(\frac{-(l_a-x-s) + \mu\theta}{\sigma\sqrt{\theta}}\right) \\ + \exp\left(\frac{2\mu(l_a-x-s)}{\sigma^2}\right) \Phi\left(\frac{-(l_a-x-s) - \mu\theta}{\sigma\sqrt{\theta}}\right). \end{aligned} \quad (31)$$

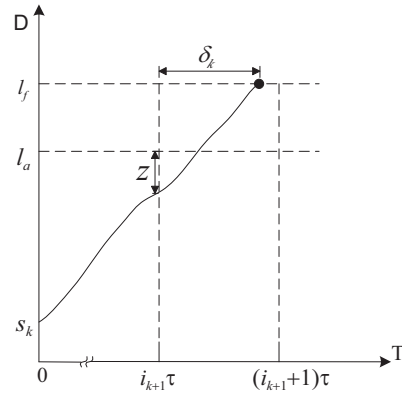


Fig. 2. Failure time δ_k in a mission after k PMs.

- 3) The detailed computation can be found in Appendix B-(2). If the system state satisfies $D(\kappa\tau) < l_a$ at the end of the κ th mission, the system is left as it is. The expected probability for such an event with respect to S_{i-1} is

$$\begin{aligned} E_{S_{i-1}}[P(D(\kappa\tau) + S_{i-1} < l_a)] = \int_0^{l_a} \left(\Phi\left(\frac{(l_a-s) - \mu\kappa\tau}{\sigma\sqrt{\kappa\tau}}\right) \right. \\ \left. - \exp\left(\frac{2\mu(l_a-s)}{\sigma^2}\right) \Phi\left(\frac{-(l_a-s) - \mu\kappa\tau}{\sigma\sqrt{\kappa\tau}}\right) \right) h_{i-1}(s) ds. \end{aligned} \quad (32)$$

5.2. Renewal cycle classification

According to the ending type of a system renewal cycle, the renewal cycle can be divided into two classes: a renewal with a CM and a renewal with a preventive replacement.

5.2.1. Renewal cycle ending with a CM

A failure can happen in any mission period before the U th mission. By summarizing all of the possibilities, we can obtain the expected operating time, the expected non-operating time and the expected cost.

Assume there are k ($k = 1, 2, \dots$) PM cycles before the CM in a renewal cycle. The inter-PM period for each PM is $(i_1+1)\tau$, $(i_2+1)\tau$, ..., $(i_k+1)\tau$, where $i_k \in \mathbf{N}$ denotes the mission number in the k th inter-PM cycle. By the time length $(i_{k+1}+1)\tau + \delta_k$ after the last PM action, the system fails and a CM action takes place immediately. $\delta_k \in (0, \tau]$ is the random failure time in a mission after the k th PM and $(i_{k+1}+1)\tau + \delta_k + \sum_{j=1}^k (i_j+1)\tau < U\tau$. According to Fig. 2, the expectation of δ_k with respect to S_k is

$$\begin{aligned} E_{S_k}[\delta_k] &= \int_0^\tau \int_0^{l_a-S_k} t f(t; l_f-l_a+z) f_{\phi}(i_{k+1}\tau; l_a-z-S_k) dz dt \\ &= \int_0^{l_a} \int_0^\tau \int_0^{l_a-s} t f(t; l_f-l_a+z) f_{\phi}(i_{k+1}\tau; l_a-z-s) h_k(s) dz dt ds, \end{aligned} \quad (33)$$

$$f(t; l_f-l_a+z) = \frac{l_f-l_a+z}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(l_f-l_a+z-\mu t)^2}{2\sigma^2 t}\right), \quad (34)$$

$$f_{\phi}(t; l_a-z-S_k) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(\frac{l_a-z-S_k-\mu t}{2\sigma^2 t}\right), \quad (35)$$

where S_k is the residual damage after the k th PM. z is a random variable, expressed as $z = l_a - D((i_k+1)\tau)$.

For a CM renewal cycle, both the number of PMs and the mission number within each PM cycle vary, and as such we have

to model all possibilities. Here we only show the formulation of the probability for the event of k PMs before a CM and with a particular combination of $i_1, \dots, i_{k+1}$. The expected operating time, non-operating time and cost given k must be the summation of all possible combinations of $i_1, \dots, i_{k+1}$. Clearly because of the availability constraint, if all of the A1 for the PM cycles before failure in a renewal cycle satisfy $A1 > \zeta$, the process is valid; otherwise, the process is invalid. The probability for such process given k PM cycles with a particular combination of $i_1, \dots, i_{k+1}$ before a CM is

$$P_{cm}(k) = P((D(i_1\tau) < l_a) \cap (l_a \leq D((i_1+1)\tau) < l_f)) \\ \times E_{S_1}[P((D(i_2\tau) + S_1 < l_a) \cap (l_a \leq D((i_2+1)\tau) + S_1 < l_f))] \\ \times \dots \times E_{S_{k-1}}[P((D(i_k\tau) + S_{k-1} < l_a) \cap (l_a \leq D((i_k+1)\tau) + S_{k-1} < l_f))] \\ \times E_{S_k}[P((D(i_{k+1}\tau) + S_k < l_a) \cap (l_a \leq D((i_{k+1}+1)\tau) + S_k < l_f))] \times \prod_{j=1}^k I_{A1 > \zeta}^{j'} \quad (36)$$

where $I_{A1 > \zeta}^{j'}$ is an indicator function, if $A1[j'] = \frac{(i_j+1)\tau}{(i_j+1)\tau + E[M_{j'}]} > \zeta$, ($j' = 1, 2, \dots, k$), $I_{A1 > \zeta}^{j'} = 1$; otherwise, $I_{A1 > \zeta}^{j'} = 0$.

Considering all of the possible $i_1, \dots, i_k$ for a renewal process involving k PM cycles, the expected operating time in a renewal cycle is

$$T_{op(1)}[k] = \sum_{i_1=0}^{U-1-k} \sum_{i_2=0}^{U-1-(i_1+1)-(k-1)} \dots \sum_{i_k=0}^{U-1-\text{Sum}(k)} \sum_{i_{k+1}=0}^{U-1-\text{Sum}(k+1)} (((i_1+1)\tau + (i_2+1)\tau + \dots \\ + (i_k+1)\tau + i_{k+1}\tau + E_{S_k}[\delta_k]) P_{cm}(k)) \quad (37)$$

where $\text{Sum}(\theta) = k - (\theta - 1) + \sum_{j=1}^{\theta-1} (i_j + 1)$ is used for simplification.

The expected non-operating time in a renewal cycle given k is

$$T_{nonop(1)}[k] = \sum_{i_1=0}^{U-1-k} \sum_{i_2=0}^{U-1-(i_1+1)-(k-1)} \dots \sum_{i_k=0}^{U-1-\text{Sum}(k)} \sum_{i_{k+1}=0}^{U-1-\text{Sum}(k+1)} ((E[M_1] + \dots + E[M_k] + \zeta) P_{cm}(k)) \quad (38)$$

The expected cost incurred in a renewal cycle given k is

$$\text{Cost}_{(1)}[k] = \sum_{i_1=0}^{U-1-k} \sum_{i_2=0}^{U-1-(i_1+1)-(k-1)} \dots \sum_{i_k=0}^{U-1-\text{Sum}(k)} \sum_{i_{k+1}=0}^{U-1-\text{Sum}(k+1)} \left(\left(C_p \sum_{j=1}^k E[M_{j'}] \right. \right. \\ \left. \left. + C_l \left(\sum_{j=1}^k (i_j + 1) + i_{k+1} \right) + C_c \right) P_{cm}(k) \right) \quad (39)$$

The final expected operating time, non-operating time and cost are the summations of Eqs. (37)–(39) with respect to k , $k=0, 1, \dots$

5.2.2. Renewal cycle ending with a preventive replacement

1) Type A preventive replacement

Type A preventive replacement means that the renewal cycle ends with a preventive replacement at $U\tau$. Assume there are k PM cycles in a renewal cycle, the inter-PM intervals for each PM are $(i_1+1)\tau$, $(i_2+1)\tau$, ..., $(i_k+1)\tau$, and $\sum_{j=1}^k (i_j+1) < U$. Before the system fails or another PM occurs, the mission sequence number reaches U . If a renewal cycle ends with type A preventive replacement, all of the A1 before the final replacement should satisfy $A1 > \zeta$. Given that $i_1, \dots, i_k$ are given, the probability for the system replaced at $U\tau$ with k PMs and $i_1, \dots, i_k$ is

$$P_{rep1}(k) = P((D(i_1\tau) < l_a) \cap (l_a \leq D((i_1+1)\tau) < l_f)) \\ \times E_{S_1}[P((D(i_2\tau) + S_1 < l_a) \cap (l_a \leq D((i_2+1)\tau) + S_1 < l_f))] \\ \times \dots \times E_{S_{k-1}}[P((D(i_k\tau) + S_{k-1} < l_a) \cap (l_a \leq D((i_k+1)\tau) + S_{k-1} < l_f))] \\ \times I_{A1 \leq \zeta}^{k+1} \prod_{j=1}^k I_{A1 > \zeta}^{j'} \quad (40)$$

$$+ S_{k-1} < l_f)) \times E_{S_k}[P(D((U-1 - \sum_{j=1}^k (i_j+1))\tau) \\ + S_k < l_a) \cap (D((U - \sum_{j=1}^k (i_j+1))\tau) + S_k < l_f))] \prod_{j=1}^k I_{A1 > \zeta}^{j'} \quad (40)$$

The expected operating time in the renewal cycle that the system is replaced at $U\tau$ with k PMs is

$$T_{op(2)}[k] = \sum_{i_1=0}^{U-1-k} \dots \sum_{i_k=0}^{U-1-\text{Sum}(k)} (U\tau P_{rep1}(k)) \quad (41)$$

The expected non-operating time with k PMs is

$$T_{nonop(2)}[k] = (E[M_1] + E[M_2] + \dots + E[M_k] + \zeta) \sum_{i_1=0}^{U-1-k} \dots \sum_{i_k=0}^{U-1-\text{Sum}(k)} P_{rep1}(k) \quad (42)$$

The expected cost with k PMs is

$$\text{Cost}_{(2)}[k] = \left(UC_l + C_p \sum_{j=1}^k E[M_{j'}] + C_R \right) \sum_{i_1=0}^{U-1-k} \dots \sum_{i_k=0}^{U-1-\text{Sum}(k)} P_{rep1}(k) \quad (43)$$

2) Type B preventive replacement

Type B preventive replacement means that the renewal cycle ends with a preventive replacement before $U\tau$ due to the availability constraint. We assume that the replacement happens at the $(k+1)$ th PM before $U\tau$. To realize the above event, the A1 over the first k PM cycles must satisfy $A1 > \zeta$, and the $(k+1)$ th PM A1 satisfies $A1 \leq \zeta$. The probability for such event given k and $i_1, \dots, i_{k+1}$ is

$$P_{rep2}(k) = P((D(i_1\tau) < l_a) \cap (l_a \leq D((i_1+1)\tau) < l_f)) \\ \times E_{S_1}[P((D(i_2\tau) + S_1 < l_a) \cap (l_a \leq D((i_2+1)\tau) + S_1 < l_f))] \\ \times \dots \times E_{S_{k-1}}[P((D(i_k\tau) + S_{k-1} < l_a) \cap (l_a \leq D((i_k+1)\tau) + S_{k-1} < l_f))] \\ \times E_{S_k}[P((D(i_{k+1}\tau) + S_k < l_a) \cap (l_a \leq D((i_{k+1}+1)\tau) + S_k < l_f))] \\ \times I_{A1 \leq \zeta}^{k+1} \prod_{j=1}^k I_{A1 > \zeta}^{j'} \quad (44)$$

where $I_{A1 \leq \zeta}^{k+1}$ is an indicator function $I_{A1 \leq \zeta}^{k+1} = 1$ when $A1[k+1] = (i_{k+1}+1)\tau / ((i_{k+1}+1)\tau + E[M_{k+1}]) \leq \zeta$; otherwise, $I_{A1 \leq \zeta}^{k+1} = 0$.

The expected operating time given k is

$$T_{op(3)}[k] = \sum_{i_1=0}^{U-2-k} \dots \sum_{i_k=0}^{U-2-\text{Sum}(k+1)} \left(P_{rep2}(k) \sum_{j=1}^{k+1} (i_j+1)\tau \right) \quad (45)$$

The expected non-operating time given k is

$$T_{nonop(3)}[k] = \sum_{i_1=0}^{U-2-k} \dots \sum_{i_k=0}^{U-2-\text{Sum}(k+1)} \left(P_{rep2}(k) \left(\sum_{j=1}^k E[M_{j'}] + \zeta \right) \right) \quad (46)$$

The expected cost given k is

$$\text{Cost}_{(3)}[k] = \sum_{i_1=0}^{U-2-k} \dots \sum_{i_k=0}^{U-2-\text{Sum}(k+1)} \left(P_{rep2}(k) \left(C_l \sum_{j=1}^{k+1} (i_j+1) + C_R \right) \right) \quad (47)$$

6. PM policy optimization

As mentioned earlier, in order to satisfy the mission requirements, the system is usually allocated with availability/reliability constraints. PM as an effective measure is used to ensure the system

operating as desired. How to develop a cost effective PM policy to achieve the goal of the mission is the focus of this section.

6.1. Optimization model

For the mission-oriented system, we choose the minimization of the long term expected maintenance cost per unit time as the objective of the proposed PM policy under mission constraints. The decision variable is the optimal PM threshold. After each renewal, the system will be restored to be as good as new. According to the renewal reward theory [30], the expected cost per unit time is equal to the ratio between the expected renewal cycle cost and the expected renewal cycle length.

The optimal PM policy can be determined by the following mathematical programming optimization:

$$\begin{aligned} \text{Min } C &= \lim_{t \rightarrow \infty} \frac{E[\text{TotalCost}(t)]}{t} = \frac{\sum_{m=1}^3 \text{Cost}(m)}{\sum_{m=1}^3 T_{op(m)} + \sum_{m=1}^3 T_{nonop(m)}} \\ \text{Subject to } &\begin{cases} A1[j'] \geq \zeta, (j' = 1, \dots, k) \\ T_{op(m)} \leq U\tau, (m = 1, 2, 3) \\ 0 < l_a < l_f. \end{cases} \end{aligned} \quad (48)$$

The first constraint ensures that the average mission availability $A1[j']$ is not lower than the threshold ζ . If $A1[j'] < \zeta$, the system will be replaced by a new one. The second constraint indicates that the total system running time should not exceed $U\tau$. When the accumulated running time is longer than $U\tau$, the system will be replaced.

The achieved long run availability is expressed as

$$A2 = \frac{\sum_{m=1}^3 T_{op(m)}}{\sum_{m=1}^3 T_{op(m)} + \sum_{m=1}^3 T_{nonop(m)}} \quad (49)$$

Since Eq. (48) uses the summation of k and the summation of the combination of $i_1, \dots, i_k$ for a given k , it is computationally intensive. However, because of the increasing nature of the residual damage and PM durations, k is always limited by the availability constraint. As such, by trial and error we can always find an upper limit for k^* and beyond which no more computing is needed.

6.2. Optimization algorithm

For solving the programming model, an optimization algorithm is designed to obtain the optimal PM policy. This algorithm searches over the range $(0, l_f]$ to determine the optimal PM threshold l_a that minimizes the cost rate with the mission constraints.

Let Type I denotes the renewal cycle ending with CM; Type II denotes the type A preventive replacement at $U\tau$; Type III denotes the type B preventive replacement due to the availability requirement. The optimization algorithm is as follows.

Algorithm for the programming model

Step 1: Start with a small value of l_a within the range $(0, l_f]$.

Step 2: Set $k=0$

Calculate $Top_{(n)}[k]$, $Tnon_{(n)}[k]$ and $Cost_{(n)}[k]$, $n=1, 2, 3$.

Step 3: Increase k by $k=k+1$

Calculate $Top_{(n)}[k] = Top_{(n)}[k-1] + Top_{(n)}[k]$,

$Tnon_{(n)}[k] = Tnon_{(n)}[k-1] + Tnon_{(n)}[k]$ and

$Cost_{(n)}[k] = Cost_{(n)}[k-1] + Cost_{(n)}[k]$, $n=1, 2, 3$.

Step 4: Calculate Eq. (48) subject to the constraint

Step 5: Repeat steps 3 and 4 until no more change is observed in Eq. (48).

Step 6: Adjust l_a by a small increment unless $l_a > l_f$ and repeat steps 2–5.

Step 7: Choose the optimal l_a with the minimization of Eq. (48).

7. A numerical example

Suppose the degradation of a system follows a Wiener process. For illustration, we consider the parameters for the Wiener-degradation process model to be $u=0.2$, $\sigma=0.02$, and $l_f=6$. The PM duration model is characterized by $\gamma=0.1$, $\lambda=0.05$, and both the preventive and CM renewal durations are assumed to be $\xi=2$. The costs for the different maintenance actions are assumed as $C_I=1$, $C_P=10$, $C_R=40$, and $C_C=80$. The threshold for the availability in a PM cycle is $\zeta=0.95$. The mission duration is $\tau=5$, and the maximum mission number is $U=7$.

Suppose that the residual damage model is characterized by $a=7$, $\varepsilon=0.4$. According to (13), the ratio of the residual damage to the threshold l_a can be listed with the increase of the PM cycle number, see Table 1. It can be seen that when the PM cycle number reaches 4, the expected residual damage will accumulate to $0.9927l_a$. If the residual damage approaches the threshold l_a too much, frequent PMs will be triggered and it will be hard to satisfy the mission availability requirement. So there will always be a limited PM cycle number in a renewal cycle.

7.1. Optimal solution

The aim of this research is to optimize the PM policy with the minimum expected cost ratio through determining the PM threshold l_a under mission constraints. For a fixed l_a , we calculate Eq. (48) with the increase of k gradually. Fig. 3 represents the contribution to the expected cost ratio with various l_a for different k , where “ $k=1$ ” in the legend means that the possible PM number appeared in the system lifecycle is 1 and so on.

From Fig. 3, we find that the cost ratios show no change when $k > 3$ for all l_a . Therefore, the up-limit for k in the calculation of

Table 1
Residual damage vs. PM number k .

PM no.	$K=1$	2	3	4
$E[S_i]/l_a$	0.5332	0.7076	0.9490	0.9927

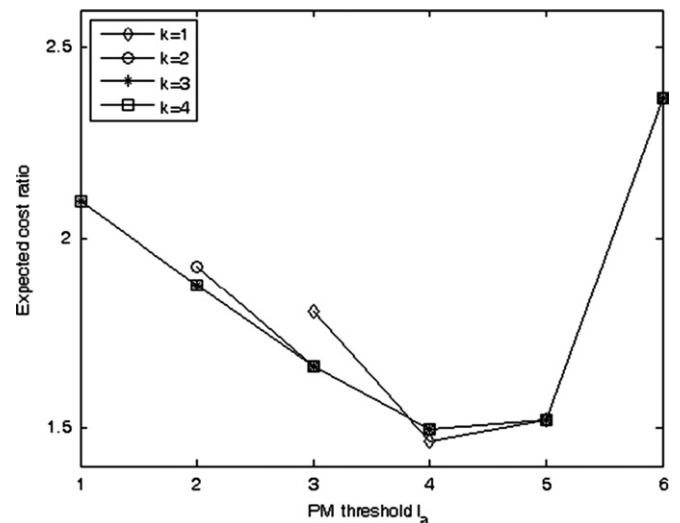


Fig. 3. Maintenance cost ratio vs. PM threshold for different k .

Eq. (48), k^* , is 3. Particularly, when $l_a > 5$, all cost ratios become identical including $k=1$. Because the residual damage after each PM and the duration occupied by the PM actions increase with

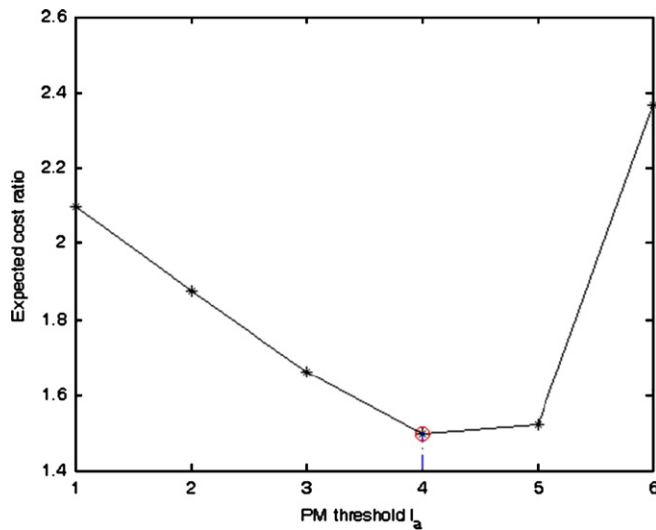


Fig. 4. Evolution of the expected cost ratio with l_a for $k^*=6$.

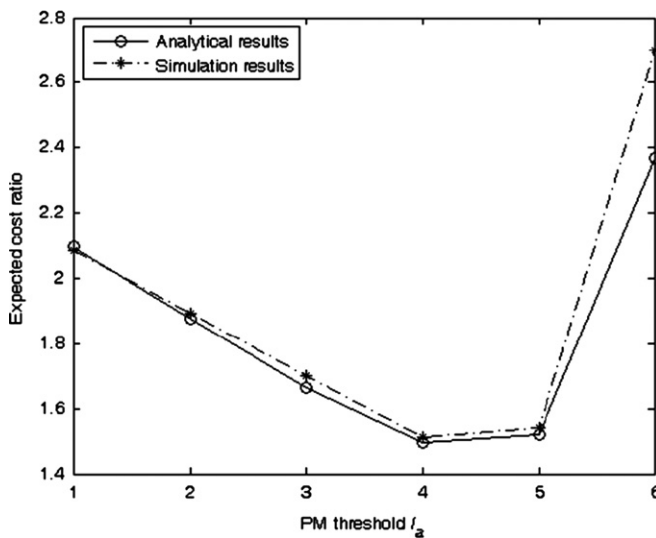


Fig. 5. Proposed analytic model vs. the simulation method.

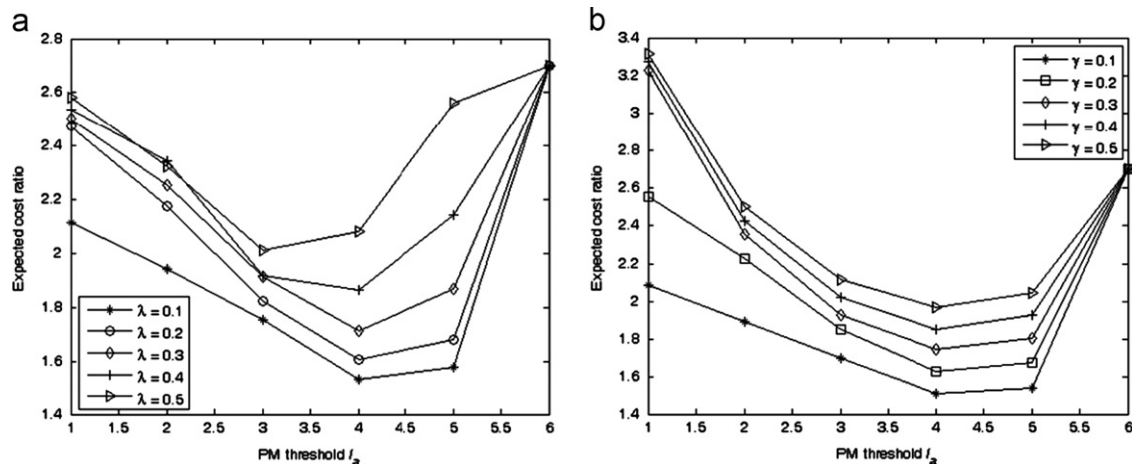


Fig. 6. Sensitivity analysis of (a) λ and (b) γ on the expected cost ratio.

the PM number based on (13) and (14), the possibility with multiple PM numbers in a renewal cycle shows a decreasing trend with the increase of l_a . In general, the cost ratio curves present a decreasing trend before $l_a=4$, because the cost for PMs decreases with l_a . Then the cost ratio increases dramatically as the failure opportunity becomes prominent with l_a .

By the optimization algorithm, the optimal cost ratio is $C^*=1.4959$, and the associated long run availability is $A_2=0.9236$, which are obtained at $l_a=4$. The optimal value is marked by a small circle in Fig. 4.

7.2. Comparison with the simulated result

Compared with the proposed model shown above, a simulation method is used to validate the correctness and accuracy of the analytical/numerical method for the mission-oriented system maintenance optimization. The simulation procedure is given in Appendix C. The simulation result is shown in Fig. 5.

From Fig. 5, we find that the cost ratio computed by the proposed method in this paper fits with the simulation result well. It is noticed that the results of the two method fits well when l_a is small, and the differences between the results of the two methods are obvious when l_a is large. The difference may be caused by the error in the numerical integration. We use the classical equally spaced steps method to compute the integral of a function. The larger l_a is, the larger spaced steps. This leads to the observed increased error of the numerical method with l_a .

7.3. Sensitivity analysis

For the mission-oriented system, the parameters of the maintenance optimization model may affect the maintenance policy in different way. This section aims to investigate the sensitivity of the maintenance policy to the parameters of the proposed maintenance optimization model. The parameters of interest include the imperfect PM duration parameters, (λ, γ) , and the availability requirement, ζ . In order to analyze the influences of the parameters to the maintenance policy, we vary one of the parameters, and fix the other parameters to investigate the evolution of the expected cost ratio.

When λ varies from 0.1 to 0.5 as shown in Fig. 6(a), the expected cost ratio increases with the increasing of λ . We can see that the minimum cost ratio increases from 1.5348 to 2.011. Fig. 6(b) shows the evolution of the expected cost ratio with γ increasing from 0.1 to 0.5. Similar to Fig. 6(a), the increasing of γ leads the minimum expected cost ratio increasing from 1.4959

to 1.9683. From (14), we know that the expected maintenance duration increases with the increasing of λ and γ . This indicates that the expected cost ratio will increase with the maintenance time increasing.

As shown in Fig. 7, when the availability requirement, ζ , increases from 0.90 to 0.98, the evolution of the expected cost ratio shows an obviously increasing trend with ζ for the small l_a , but not obvious for the large l_a . Because the PM duration increases with the l_a quickly and makes the average mission availability very low for the large value of l_a . The mission almost only allows one PM taken on the system when l_a is large so the cost ratios are almost the same for different ζ , i.e., the expected cost ratio is insensitive to ζ when l_a exceeds 4 in Fig. 7. For this case, we notice that when the PM number is small, the influence of ζ is obvious, otherwise, the influences to the expected cost ratio is weak. This also indicates the ζ can effectively control the PM number.

The parameters of γ , λ and ζ have different influences to the maximum PM number k^* . From Fig. 8, we can see that when γ varies from 0.1 to 1.0 or λ increases from 0.1 to 2.0, k^* decreases from 3 to 1. When ζ increases from 0.88 to 0.99, k^* declines from

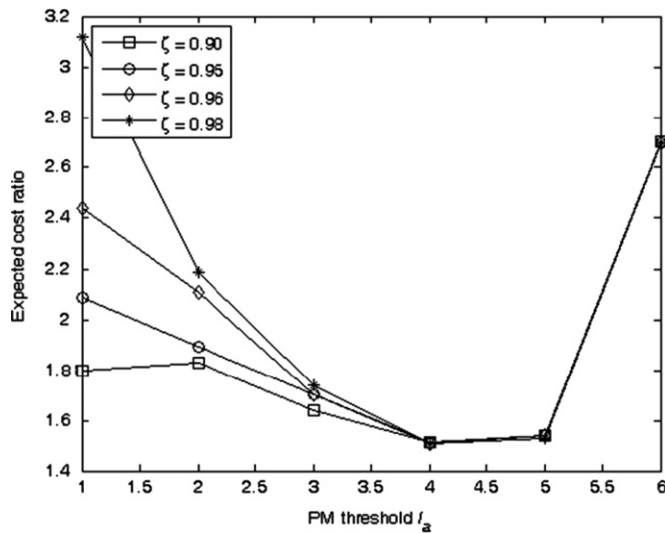


Fig. 7. Sensitivity analysis of ζ on the expected cost ratio.

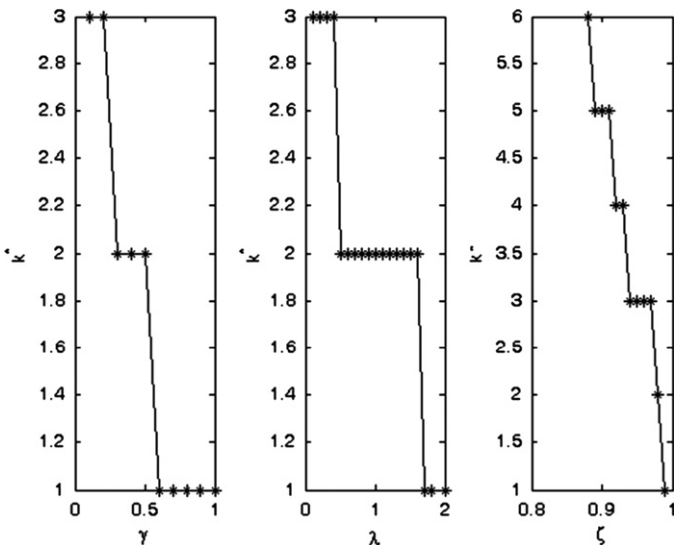


Fig. 8. Influences of varied γ , λ and ζ on k^* .

6 to 1. Compared with the influences of λ and γ , ζ has a greater influence on k^* . From the sensitivity analysis, we can conclude that the mission requirements have important influences on the maintenance policy, and we need to consider the mission requirements during the maintenance optimization.

8. Conclusions

In this paper, a CBM model to optimize the PM policy for a mission-oriented system is developed based on a Wiener-degradation process. Most of the existing CBM models focused on the minimization of maintenance cost or the maximization of the system availability/reliability with perfect PM and very few considered the mission constraints and the imperfect PM nature. Compared with the existing CBM optimization models, the proposed model takes into account the mission constraints, such as mission availability requirement under the assumption of imperfect PM, and makes the maintenance model more practical for the mission oriented system. By classifying the renewal cycles, we have studied the stochastic characteristics of the mission-oriented system and developed a maintenance optimization model with the minimization of the maintenance cost as the objective. For obtaining the optimal results, a searching algorithm is used to obtain the optimal PM threshold. The numerical example shows that the mission constraints have considerable impact on the optimization result and proves the usefulness of the proposed model for the mission-oriented system.

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Appendix A

$E[S_i] = c_i l_a \left[1 - \frac{1}{\varepsilon a^{i-1}} (1 - \exp(-\varepsilon a^{i-1})) \right]$ with $c_i = 1 / (1 - \exp(-\varepsilon a^{i-1}))$, $a > 1$, $\varepsilon > 0$ is an increasing function with PM number i increasing.

Proof. The derivative of $E[S_i]$ with respect to i can be expressed as

$$\begin{aligned} \frac{\partial E[S_i]}{\partial i} &= \partial l_a \left(\frac{1}{1 - \exp(-\varepsilon a^{i-1})} - \frac{1}{\varepsilon a^{i-1}} \right) / \partial i \\ &= l_a \left(\frac{-\varepsilon a^{i-1} \ln a \exp(-\varepsilon a^{i-1})}{(1 - \exp(-\varepsilon a^{i-1}))^2} + \frac{\ln a}{\varepsilon a^{i-1}} \right) \\ &= l_a \ln a \left(\frac{-\varepsilon a^{i-1} \exp(\varepsilon a^{i-1})}{(\exp(\varepsilon a^{i-1}) - 1)^2} + \frac{1}{\varepsilon a^{i-1}} \right). \end{aligned} \quad (A.1)$$

If we denote $A = \varepsilon a^{i-1} > 0$, then

$$\frac{\partial E[S_i]}{\partial i} = l_a \ln a \left[\frac{-A \exp(A)}{(\exp(A) - 1)^2} + \frac{1}{A} \right] = l_a \ln a \frac{-A^2 \exp(A) + (\exp(A) - 1)^2}{A(\exp(A) - 1)^2} \quad (A.2)$$

For $A > 0$, we can obtain $A(\exp(A) - 1)^2 > 0$.

For $a > 1$, $l_a > 0$, $\frac{\partial E[S_i]}{\partial i} > 0$ holds if $-A^2 \exp(A) + (\exp(A) - 1)^2 > 0$.

Define $G[A] = -A^2 \exp(A) + (\exp(A) - 1)^2$, and we have

$$\begin{aligned} \frac{\partial G[A]}{\partial A} &= 2\exp(2A) - 2\exp(A) - 2A\exp(A) - A^2\exp(A) \\ &= \exp(A)(2\exp(A) - 2 - 2A - A^2). \end{aligned} \quad (A.3)$$

Through series expansion, we can get

$$\exp(A) = 1 + \frac{A}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots + \frac{A^n}{n!} + \cdots \quad (\text{A.4})$$

Then we have

$$2\exp(A) - 2 - 2A - A^2 = 2 \left(\frac{A^3}{3!} + \frac{A^4}{4!} + \cdots + \frac{A^n}{n!} + \cdots \right) > 0 \quad (\text{A.5})$$

Thus $\partial G[A]/\partial A > 0$, $G[A]$ is a monotonic increasing function.

According to $G[0]=0$ and $A > 0$, we can obtain

$$G[A] > 0 \Rightarrow \frac{\partial E[S_i]}{\partial i} > 0$$

So $E[S_i]$ shows an increasing trend with the PM number increasing. $\square$

Appendix B

1. The expected probability for a CM occurring within the κ th mission for the i th PM cycle with respect to S_{i-1} . Based on the independent increments property of the Wiener process, we have

$$\begin{aligned} E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) + S_{i-1} \geq l_f)] \\ = E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) - D((\kappa-1)\tau) \\ + S_{i-1} \geq l_f - D((\kappa-1)\tau))] \\ = E_{S_{i-1}}[P(D((\kappa-1)\tau) < l_a - S_{i-1} \cap D(\tau) \geq l_f - D((\kappa-1)\tau) - S_{i-1})] \\ = \int_0^{l_a} \int_0^{l_a-s} \int_0^\tau \int_{(\kappa-1)\tau}^\infty f_{T_a}(t|s) f_{l_f-x}(\theta; x|s) f_\phi(x; (\kappa-1)\tau) \\ h_{i-1}(s) dt d\theta dx ds \\ = \int_0^{l_a} \int_0^{l_a-s} (1 - F_{T_a}((\kappa-1)\tau|s)) F_{l_f-x}(\tau; x|E[S_{i-1}]) \\ f_\phi(x; (\kappa-1)\tau) h_{i-1}(s) dx ds, \end{aligned} \quad (\text{B.1})$$

where

$$f_{T_a}(t|s) = \frac{l_a-s}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(l_a-s-\mu t)^2}{2\sigma^2 t}\right) \quad (\text{B.2})$$

$$\begin{aligned} F_{T_a}(t|s) = P(T_a < t|s) = \Phi\left(\frac{-(l_a-s)+\mu t}{\sigma\sqrt{t}}\right) + \exp\left(\frac{2\mu(l_a-s)}{\sigma^2}\right) \\ \Phi\left(\frac{-(l_a-s)-\mu t}{\sigma\sqrt{t}}\right) \end{aligned} \quad (\text{B.3})$$

$$f_{l_f-x}(\theta; x|s) = \frac{l_f-x-s}{\sqrt{2\pi\sigma^2\theta^3}} \exp\left(-\frac{(l_f-x-s-\mu\theta)^2}{2\sigma^2\theta}\right) \quad (\text{B.4})$$

$$\begin{aligned} F_{l_f-x}(\theta; x|s) = P(T_{l_f-x} < \theta|s) \\ = \Phi\left(\frac{-(l_f-x-s)+\mu\theta}{\sigma\sqrt{\theta}}\right) + \exp\left(\frac{2\mu(l_f-x-s)}{\sigma^2}\right) \Phi\left(\frac{-(l_f-x-s)-\mu\theta}{\sigma\sqrt{\theta}}\right). \end{aligned} \quad (\text{B.5})$$

$$f_\phi(x; t) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{(x-\mu t)^2}{2\sigma^2 t}\right) \quad (\text{B.6})$$

(B.2)–(B.5) can be obtained based on (16) and (17).

2. The probability for a PM at the end of the κ th mission for the i th PM cycle with respect to S_{i-1} .

$$\begin{aligned} E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap l_a \leq D(\kappa\tau) + S_{i-1} < l_f)] \\ = E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) + S_{i-1} \geq l_a)] \\ - E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) + S_{i-1} \geq l_f)] \\ = E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) - D((\kappa-1)\tau) \\ + S_{i-1} \geq l_a - D((\kappa-1)\tau))] \end{aligned}$$

$$\begin{aligned} - E_{S_{i-1}}[P(D((\kappa-1)\tau) + S_{i-1} < l_a \cap D(\kappa\tau) - D((\kappa-1)\tau) \\ + S_{i-1} \geq l_f - D((\kappa-1)\tau))] \\ = E_{S_{i-1}}[P(D((\kappa-1)\tau) < l_a - S_{i-1} \cap D(\tau) \geq l_a - D((\kappa-1)\tau) - S_{i-1})] \\ - E_{S_{i-1}}[P(D((\kappa-1)\tau) < l_a - S_{i-1} \cap D(\tau) \geq l_f - D((\kappa-1)\tau) - S_{i-1})] \\ = \int_0^{l_a-s} \int_0^\tau \int_{(\kappa-1)\tau}^\infty f_{T_a}(t|s) f_{l_f-x}(\theta; x|s) f_\phi(x; (\kappa-1)\tau) dt d\theta dx \\ - \int_0^{l_a-s} \int_0^\tau \int_{(\kappa-1)\tau}^\infty f_{T_a}(t|s) f_{l_f-x}(\theta; x|s) f_\phi(x; (\kappa-1)\tau) dt d\theta dx \\ = (1 - F_{T_a}((\kappa-1)\tau|s)) \int_0^{l_a-s} (F_{l_a-x}(\tau; x|s) - F_{l_f-x}(\tau; x|s)) \\ f_\phi(x; (\kappa-1)\tau) dx, \end{aligned} \quad (\text{B.7})$$

where

$$f_{l_a-x}(\theta; x|s) = \frac{l_a-x-s}{\sqrt{2\pi\sigma^2\theta^3}} \exp\left(-\frac{(l_a-x-s-\mu\theta)^2}{2\sigma^2\theta}\right) \quad (\text{B.8})$$

$$\begin{aligned} F_{l_a-x}(\theta; x|s) = P(T_{l_a-x} < \theta|s) = \Phi\left(\frac{-(l_a-x-s)+\mu\theta}{\sigma\sqrt{\theta}}\right) \\ + \exp\left(\frac{2\mu(l_a-x-s)}{\sigma^2}\right) \Phi\left(\frac{-(l_a-x-s)-\mu\theta}{\sigma\sqrt{\theta}}\right). \end{aligned} \quad (\text{B.9})$$

Appendix C

The simulation algorithm for Section 7.2 is as follows.

Begin

Initialize $U, \tau, l_f, \mu, \sigma, \zeta, \zeta, C_l, C_p, C_R, C_C$

Do while ($l_a \leq l_f$)

Initial configuration Num , which is the repeat simulation times for a fixed l_a ;

Generate a null array $Temp$ with $1 \times Num$ dimensions for saving results.

Do $i=1$ to Num

Configure $PMNum=0, Runtime=0, Downtime=0, PMPoint=0, Cost=0, TempRun=0, TempDown=0$;

Do while ($t \leq U\tau + 1$)

If $t \geq U\tau$ **then**

$Runtime=U\tau$;

$Downtime=Downtime+\xi$;

$Cost=Cost+C_R$; **break**;

End

If $S(l_a, PMNum) + D(t - PMPoint) \geq l_f$ **Then**

$Runtime=t$;

$Downtime=Downtime+\xi$;

$Cost=Cost+C_C$; **break**;

End

$TempRun=t - PMPoint$

$TempDown=PMtime(l_a, PMNum)$;

If t is the inspection time **then**

If $l_a \leq S(l_a, PMNum) + D(t - PMPoint) < l_f$ **then**

If $TempRun/TempRun + TempDown \leq A1$ **then**

$Downtime=Downtime+\xi$;

$Runtime=t$;

$Cost=Cost+C_R+C_i$; **break**;

Else if $TempRun/TempRun + TempDown > A1$ **Then**

$PMNum=PMnum+1$;

$PMPoint=t$,

$Downtime=Downtime+PMtime(l_a, PMNum)$;

$Cost=Cost+C_p PMtime(l_a, PMNum)+C_i$;

End

End

If $S(l_a, PMNum) + D(t - PMPoint) < l_a$ **then**

Runtime = *t*;

Cost = *Cost* + *C_i*;

End

End

End

Temp(*i*) = *Cost* / (*Runtime* + *Downtime*)

End

Output: $\left[l_a, \frac{1}{Num} \sum_{i=1}^{Num} Temp(i) \right]$

End

Note: *Runtime* and *Downtime* are the accumulated run time and down time, respectively. *Cost* is the accumulated cost. *PMNum* is the accumulated PM number. *PMPoint* denotes the time of PM occurrence. $S(l_a, PMNum)$ is the random number function of residual damage, which is can be obtained through Eq. (12). $PMtime(l_a, PMNum)$ is the time function for the PM action. *TempRun* and *TempDown* are the temporary variables.

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